

PROJECT REPORT ON

“Real Estate Management System”



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Abstract

The Real Estate Management System is a web-based platform that simplifies buying and selling properties. It enables property owners to list their properties and allows potential buyers to search, filter, and express interest in available listings.

Key features include user authentication, property listings, advanced search filters, booking requests, and administrative controls.

The system uses React for the frontend, Node.js with Express for the backend, and MySQL as the database. For secure user authentication, it leverages JSON Web Tokens (JWT).

The primary goal of this project is to deliver a smooth, user-friendly experience for buyers and administrators, while maintaining security and responsiveness across the platform.

This project showcases how different modern technologies can be integrated to build a fully functional web application with practical, real-world use.

1. Introduction

1.1 Purpose

The **Real Estate Management System** is a web-based platform that facilitates property listings, user interactions, and transactions between buyers and sellers. The system provides a seamless and efficient way for users to browse, list, and manage real estate properties.

1.2 Scope

This system will allow property owners to list their buyers to search and express interest. The backend will handle data storage, authentication, and business logic, while the frontend will provide an interactive user interface.

2. Objective

The **Real Estate Management System** aims to create a website where property owners can list their properties and buyers can effortlessly search, view, and show interest. The system will make it easier for users to manage property details, connect, and complete real estate transactions.

3. Technology Used

- **Frontend:** React (JavaScript, HTML, CSS, TailwindCSS)
- **Backend:** ExpressJS (Node.js)
- **Database:** MySQL
- **Authentication:** JWT-based authentication

4. Requirements Analysis

4.1 Functional Requirements Analysis

Requirement	Description
User Authentication	Users (buyers admins) can register, log in, and securely access the platform using JWT-based authentication.
User Role Management	Role-based access control is implemented to differentiate between Admin and Buyer capabilities.
Property Management	Users (Admins/Sellers) can create, update, delete, and manage property listings.
Property Search & Filter	Buyers can search and filter properties based on various criteria like location, price, property type, and availability

Contact Seller	Buyers can view seller contact details or initiate contact to express interest in a property.
Favorites/Wishlist	Buyers can save properties to a wishlist for easy access later
Admin Dashboard	Admins manage the access to a dashboard for monitoring and managing users and property listings.

4.2 Non - Functional Requirements Analysis

Category	Requirement
Security	Implementation of secure database connections and encrypt sensitive user data. JWT is used for session management.
Scalability	The system is scalable to support many users and property listings without degrading performance.
Performance	Page loads, searches, and actions (like login or property save) occur within acceptable time limits.

Usability	The user interface is intuitive, responsive, and visually consistent.
Reliability	The system handles edge cases and errors gracefully without crashing or losing data.
Maintainability	The system architecture is modular and easy to update or extend (e.g., adding new roles or features).

5. Methodology Used

This project was developed using a **phased development approach** that combines elements of the **Waterfall Model** with **basic Agile practices**. As a student project, the focus was on simplicity, clarity, and learning throughout the process. The main stages of development were as follows:

5.1 Requirement Analysis

- Identified core features and functionality needed for the Real Estate Management System.
- Defined user roles (Admin, Buyer) and mapped out key use cases like property listing, search, and booking.

5.2 System Design

- Designed the database schema with three main tables: Users, Properties, and Bookings.

- Created wireframes and planned the layout for the front end using React and CSS.
- Decided on key technologies: React for the front end, Node.js with Express for the backend, MySQL for data storage, and JWT for authentication.

5.3 Implementation

- Built the frontend components using React and styled them with CSS for a responsive UI.
- Developed backend APIs with ExpressJS to handle authentication, property management, and booking operations.
- Integrated the front end and back end with RESTful API calls.

5.4 Testing

- Performed manual testing for each module (user login, property listing, search filters, etc.).
- Checked for functionality, form validation, and responsiveness across devices.

6. Implementation:

1. User Authentication

- Users log in via a web form, and credentials are verified against the database.

- After successful authentication, users are directed to their respective dashboards (**Admin** or **Buyer**).

2. Property Management

- Property owners add listings through the web interface, which updates the database in **MySQL**.
- Property details (descriptions, images, seller information) are stored and displayed on the platform for buyers to view.
- **Admin Controls:**
 - The admin panel interacts directly with the database to **add, update, or delete** property listings.
 - Admins can also manage **user permissions and roles**.

3. Searching and Filtering

- Buyers use search filters (location, price, property type, etc.) in the UI to find properties.
- Direct database queries retrieve and display relevant results based on user-selected filters.

4. Booking and Scheduling

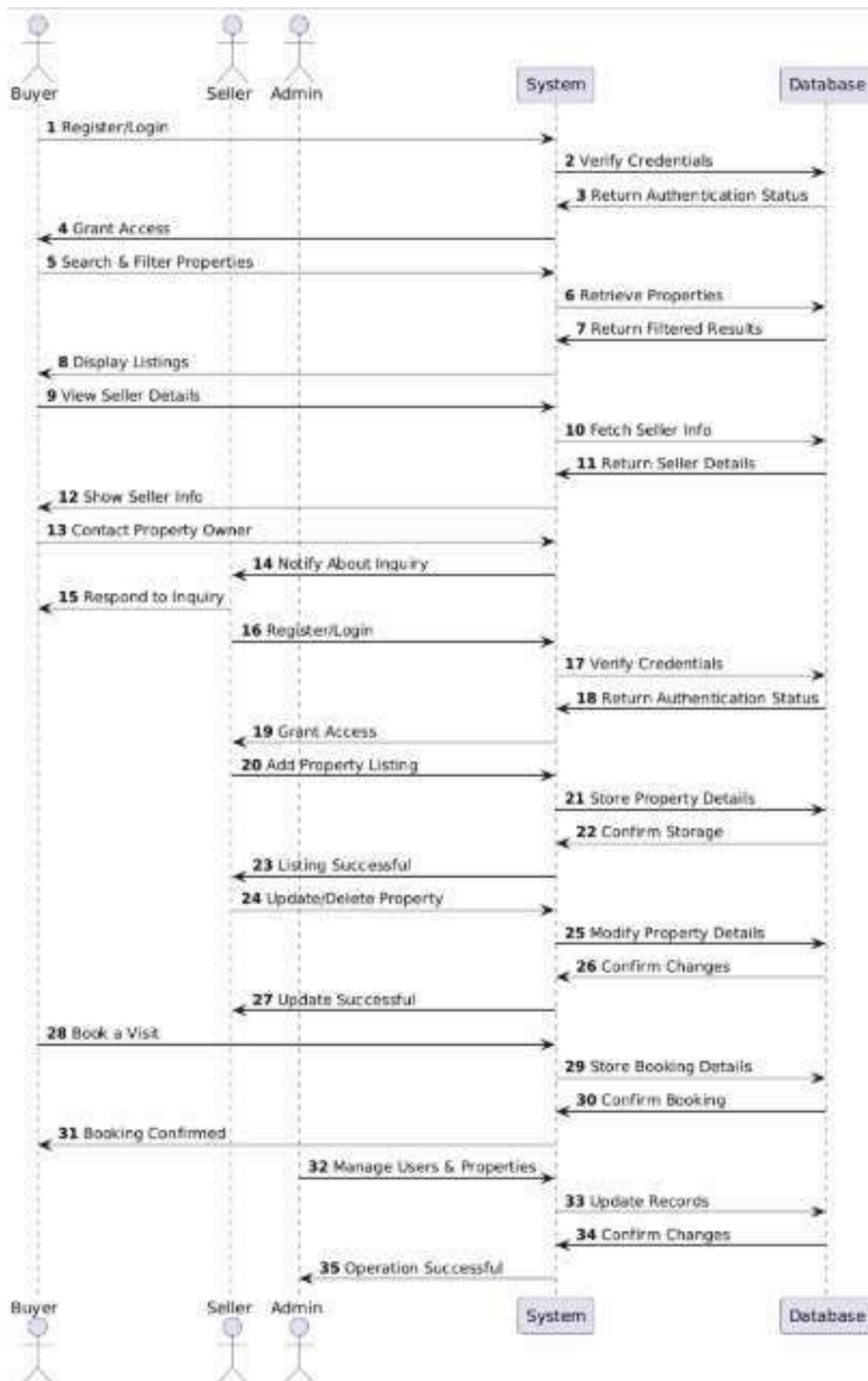
- Buyers can express interest, submit booking requests, or schedule viewings.
- These requests are **inserted into the database** and displayed via the web interface for sellers/admins to manage.

5. Rating and Feedback

- After completing a property booking, buyers are prompted to **rate the property** and provide feedback.
- Ratings and feedback are stored in the database and displayed on the property listing page for future buyers.

6. Property Details & Seller Information

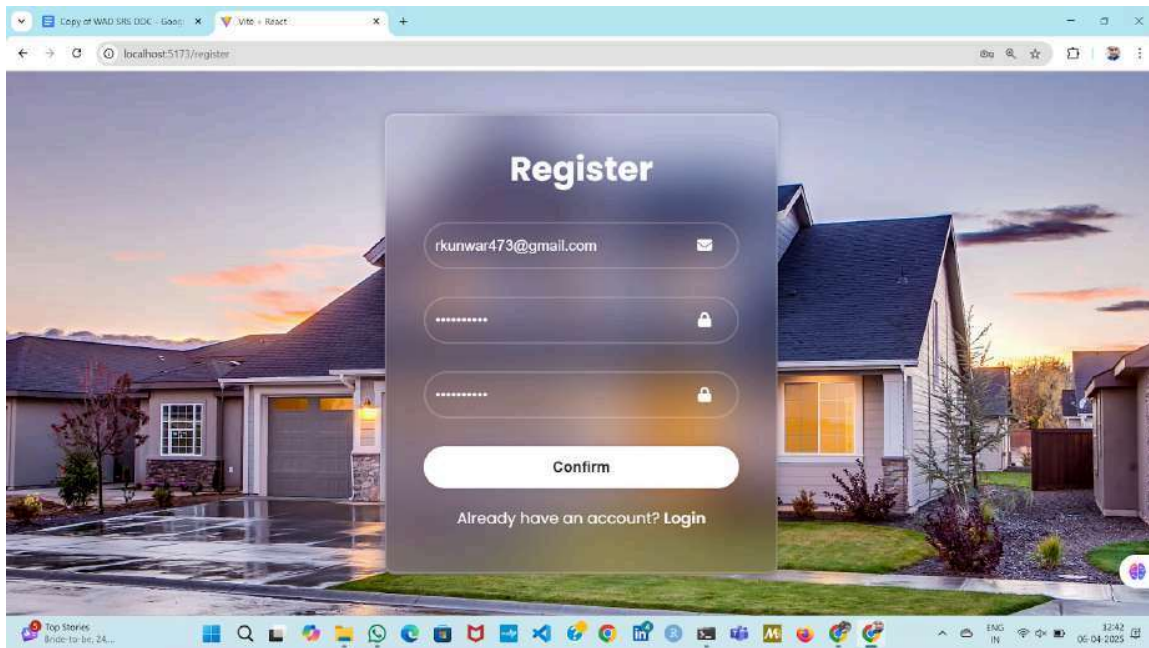
- Buyers can view **detailed property information**, including:
 - High-quality images
 - Property descriptions
 - Seller contact details
- This helps buyers make informed decisions before booking or expressing interest.



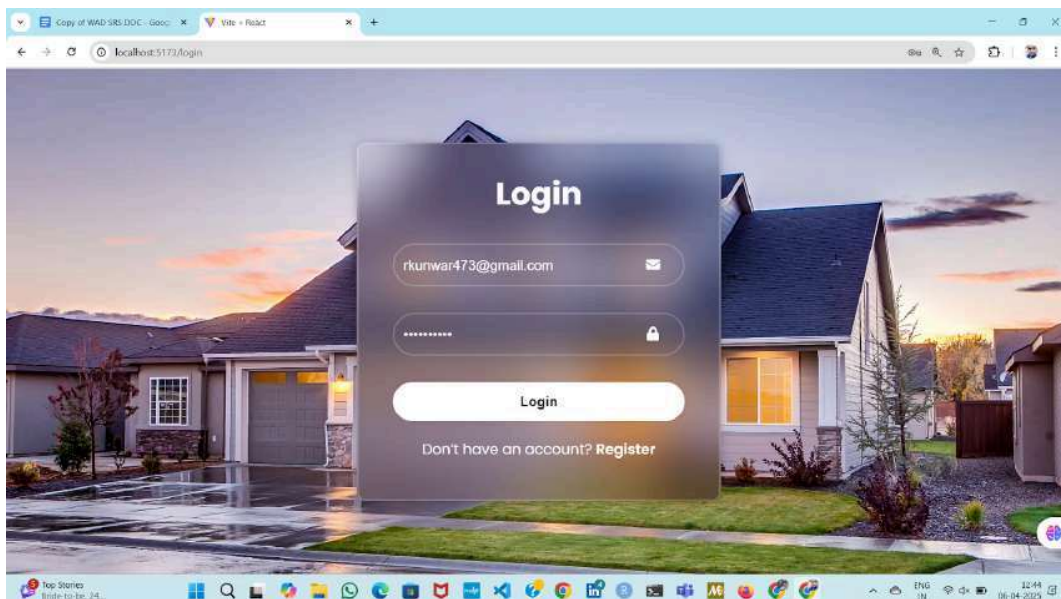
7. Results

Front-End:

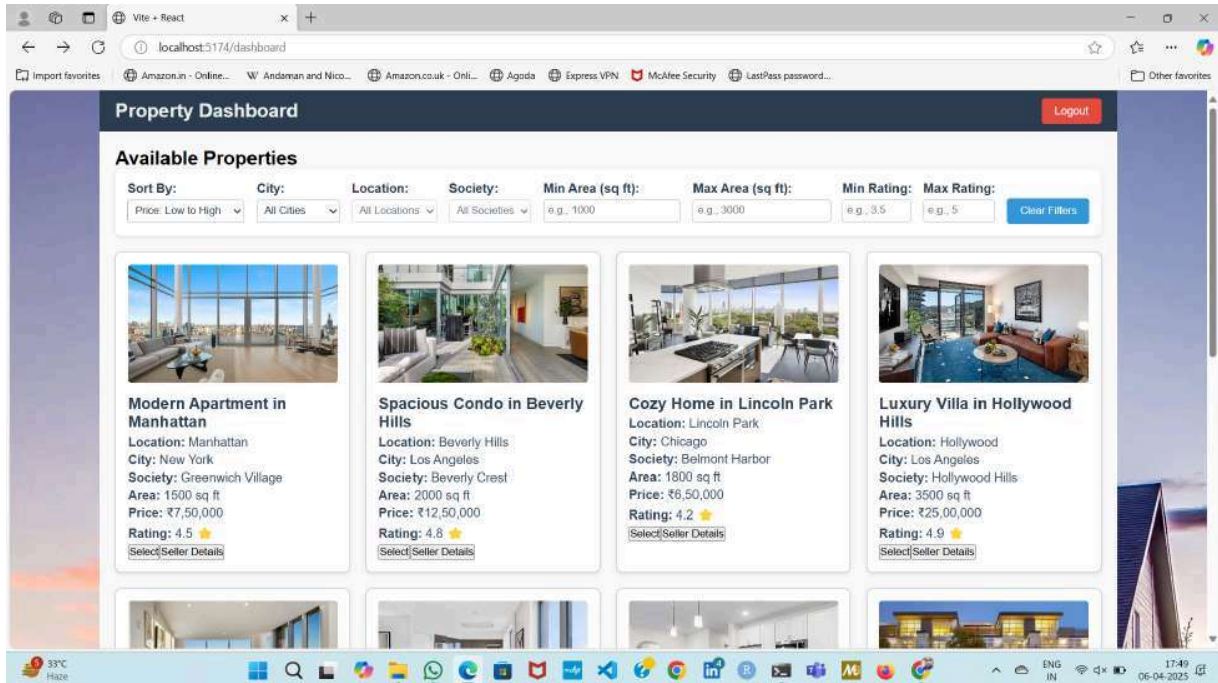
1. Registration page:



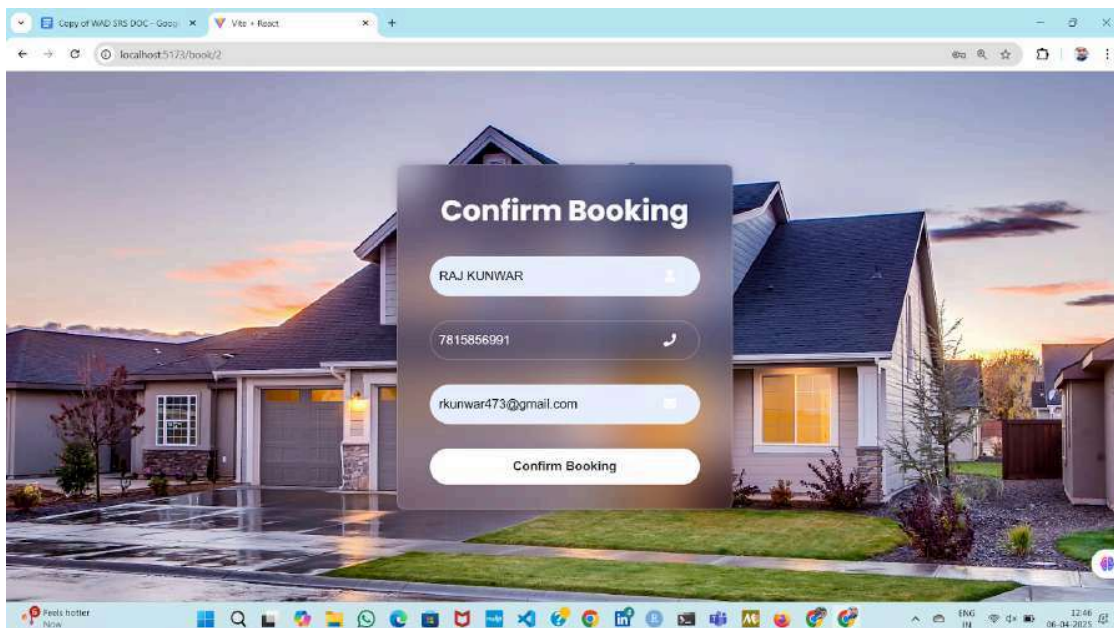
2. Login Page:



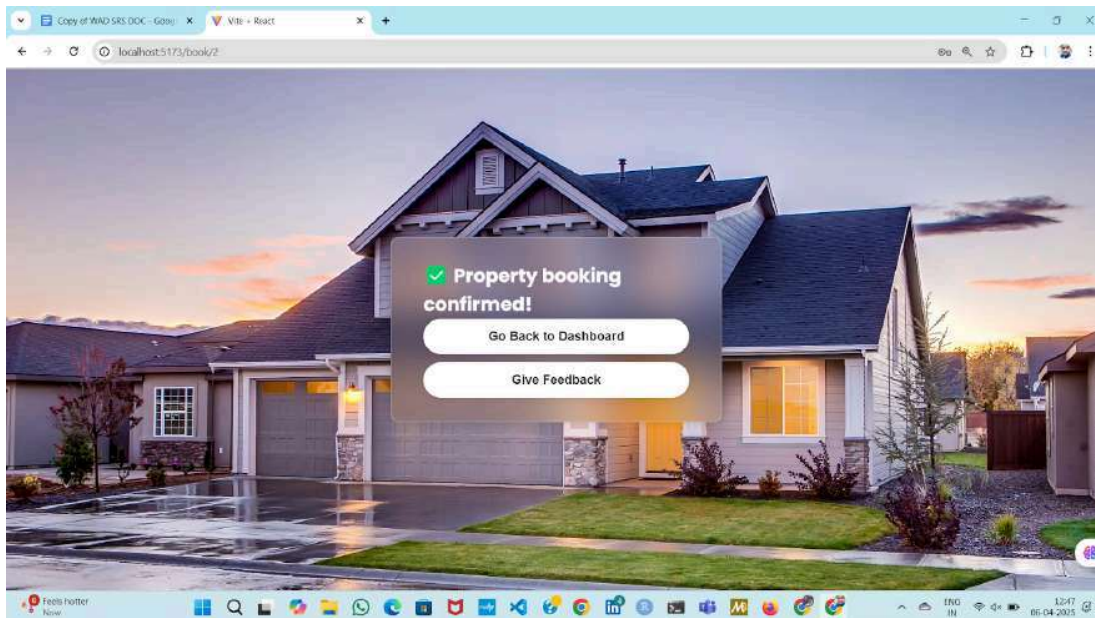
3. Dashboard:



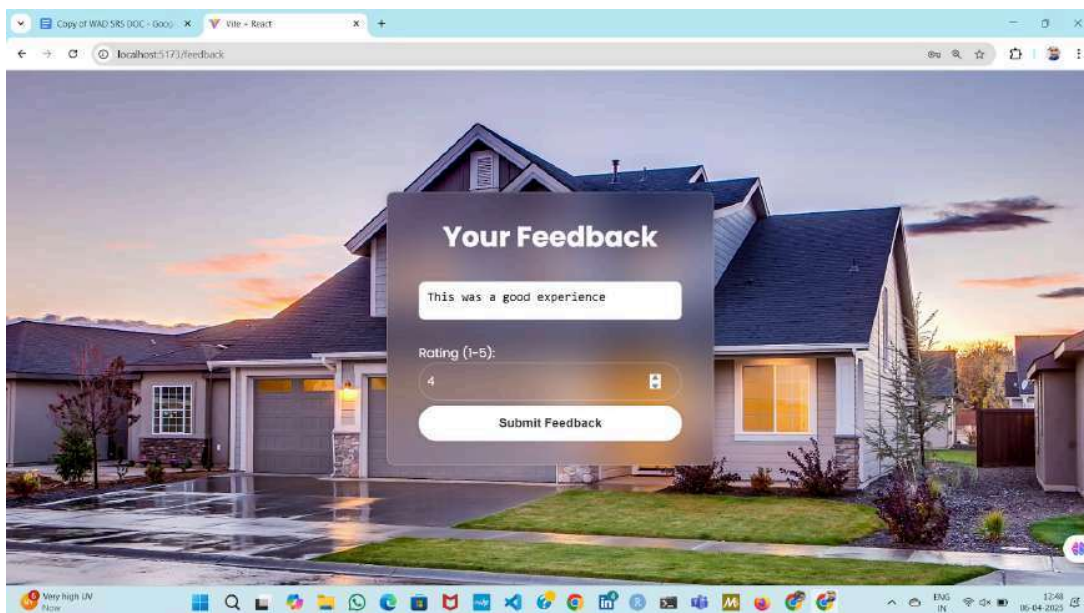
4. Booking Confirmation Page :



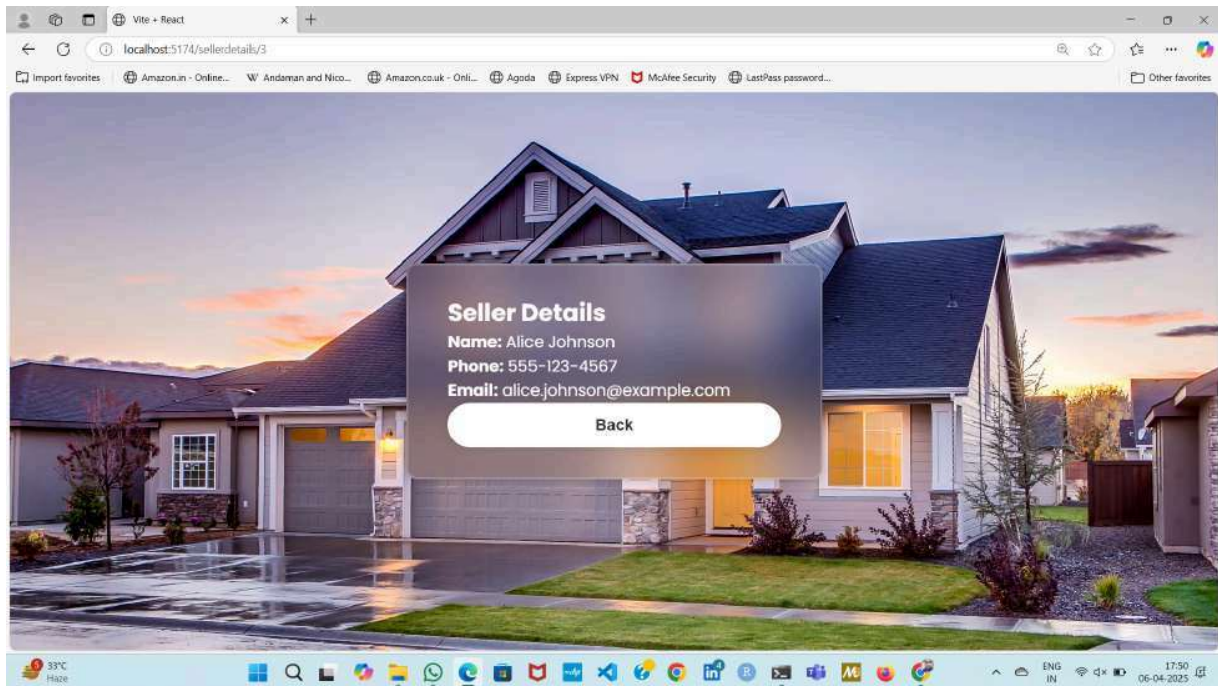
5. Back To Dashboard Page:



6. Feedback page:



7. Seller Details Page:



Back-End:

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.22631.5039]
(c) Microsoft Corporation. All rights reserved.

D:\E-state\E-state\E-state - Copy\Server>node index.js
[✓] Connected to MySQL database
[✓] Server running on http://localhost:5000
[✓] Feedback table ready!
```

```
mysql> select * from users;
Empty set (0.00 sec)

mysql> select * from users;
+-----+-----+-----+-----+
| id | EmailID | Password | Role |
+-----+-----+-----+-----+
| 3 | rkunwar473@gmail.com | $2b$10$gYladachHILvQ7B0pCUjo0tgtylw18o.QzLo0Jler5LdLLpH3xS/q | user |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> select * from bookings;
+-----+-----+-----+-----+-----+-----+
| id | user_email | user_name | user_phone | property_id | timestamp |
+-----+-----+-----+-----+-----+-----+
| 6 | rkunwar473@gmail.com | RAJ KUNWAR | 7815856991 | 2 | 2025-04-06 12:47:28 |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.01 sec)

mysql> select * from feedbacks;
Empty set (0.00 sec)

mysql> select * from feedbacks;
+-----+-----+-----+-----+
| id | feedback_message | rating | created_at |
+-----+-----+-----+-----+
| 2 | This was a good experience | 4 | 2025-04-06 12:50:42 |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> |
```

8. Conclusion

Working on the **Real Estate Management System** has been a valuable and enriching learning experience. The project allowed me to apply front-end and back-end development skills to build a fully functional web application that simplifies property listing, searching, and booking.

I gained hands-on experience with modern technologies like React, Node.js, Express, and MySQL throughout the development process. I also learned how to implement secure authentication using JWT, structure a relational database, and create RESTful APIs to connect the front end and back end effectively.

This project helped me understand the importance of planning, breaking tasks into modules, and testing at every stage to ensure a smooth user experience. I faced and solved real-world challenges such as handling form validation, data flow between components, and managing user roles.

This system can be enhanced in the future with features like payment integration, real-time messaging between buyers and sellers, and deployment to a live server. Overall, this project strengthened my technical knowledge and improved my problem-solving, time management, and documentation skills.

9. Future Scope

- AI-based property recommendations.
- Integration with third-party payment gateways.
- Virtual property tours using 3D visualization.
- Mobile application for better accessibility.

10. Reference:

Front End:

1. <https://developer.mozilla.org/en-US/docs/Web/JavaScript/React>
 - This resource provided comprehensive documentation on React, which was used for the front-end development of the Real Estate Management System.

Back End:

1. <https://nodejs.org/en/docs/>
 - The official documentation of Node.js was used to guide the system's backend development.
2. <https://expressjs.com/en/starter/installing.html>
 - This source helped set up and configure Express as the backend framework for the project.
3. <https://dev.mysql.com/doc/>
 - The official MySQL documentation was used to design and interact with the database to store property listings, user details, and transactions.

General:

1. <https://www.w3schools.com>
 - W3Schools offered practical examples and tutorials on HTML, CSS, and JavaScript, which were beneficial for front-end development.

To Troubleshoot issues:

1. <https://stackoverflow.com>
 - Stack Overflow was an invaluable resource for troubleshooting issues and seeking help during development.